



Design Task - Lorry Crane Hydraulics

REPORT FOR DESIGN TASK IN FLUID POWER SYSTEMS

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Abstract: The following report consists of the design and mathematical process which we underwent to design the hydraulic circuit for a lorry crane. The system contains two cylinder, and it was our task to determine the operating conditions for the cylinders and pump, and pick-out the appropriate components for the task. The dimensioning also required us to consider a work-case scenario, which informed our decision during part-selection.

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1. Design Task Definition

The objective of the design is to design a lorry crane, that can operative with a given mass over a defined drive cycle, and withstand a worst-case loading scenario.

For the task a certain values were given, mainly defining the drive cycle. Some values were true for all group, while others were group specific.

1.1. General Values

The general values for the drive-cycle provided included:

- $v_{p01} \rightarrow$ Normalised velocity vector for piston 1
- $v_{p02} \rightarrow$ Normalised velocity vector for piston 2
- $e_1 \rightarrow$ Level-arm for cylinder 1
- $e_2 \rightarrow$ Level-arm for cylinder 2
- $\theta_1 \rightarrow$ First arm angle
- $\theta_2 \rightarrow$ Second arm angle
- $L_0 \rightarrow$ Base height for lorry crane
- $p_2 \rightarrow$ Pressure on low-pressure side
- $\eta \rightarrow$ Oil dynamic viscosity
- $c_v \rightarrow$ Laminar leakage loss coefficient
- $k_p \rightarrow$ Coloumb friction coefficient
- $k_v \rightarrow$ Viscous friction loss coefficient
- $k_\epsilon \rightarrow$ Displacement coefficient

The information on the work-case scenario, provided the following values:

Parameter	Cylinder 1	Cylinder 2
$\theta_{1,wc}[\text{rad}]$	1,2	0,35
$\theta_{2,wc}[\text{rad}]$	0,0	0,0
$e_{1,wc}[\text{m}]$	0,14	0,48
$e_{2,wc}[\text{m}]$	0.57	0,17

1.2. Group-specific values

There were many group-specific values that we were given for the design task, these included:

- $M_{t,wc} \rightarrow$ Maximum total load mass
- $q_{r,wc} \rightarrow$ Worst-case flow to drive-cycle-max flow ratio
- $M_t \rightarrow$ Total load mass
- $L_1 \rightarrow$ First arm length
- $L_2 \rightarrow$ Second arm length
- $v_{p0} \rightarrow$ Reference cylinder velocity
- $t_0 \rightarrow$ Drive-cycle Time
- $P_{s,max} \rightarrow$ Maximum system pressure
- $\eta_c \rightarrow$ Cylinder Efficiency

1.3. Denormalisation and discretization

From the provided values we can see that the velocity profile which we were provided has been normalised and we have been provided a reference value. Using this reference value we denormalised the normalised velocity profile, to determine the effective velocity profile of the two cylinders.

$$v_{p1} = v_{p0} \cdot v_{p01} v_{p2} = v_{p1} \cdot v_{p02}$$

Additionally, we discretized time to a number of discrete times equal to the length of the provided vectors for velocity, angles and arms, as such:

```
time = linspace(0,t_0,length(e_1));
```

2. Cylinder Sizing and Selection

2.1. Crane Kinematics

Before finding the forces on the cylinder we have to determine the kinematics of the crane tip throughout the drive-cycle. We found the x and z position of the crane tip with the following equations:

$$x = L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_2)$$
$$z = L_1 \cdot \sin(\theta_1) + L_2 \cdot \sin(\theta_2) + L_0$$

Using these two values we can create two graphs, one of the tip height through time and one of the position of the crane in space through the cycle.

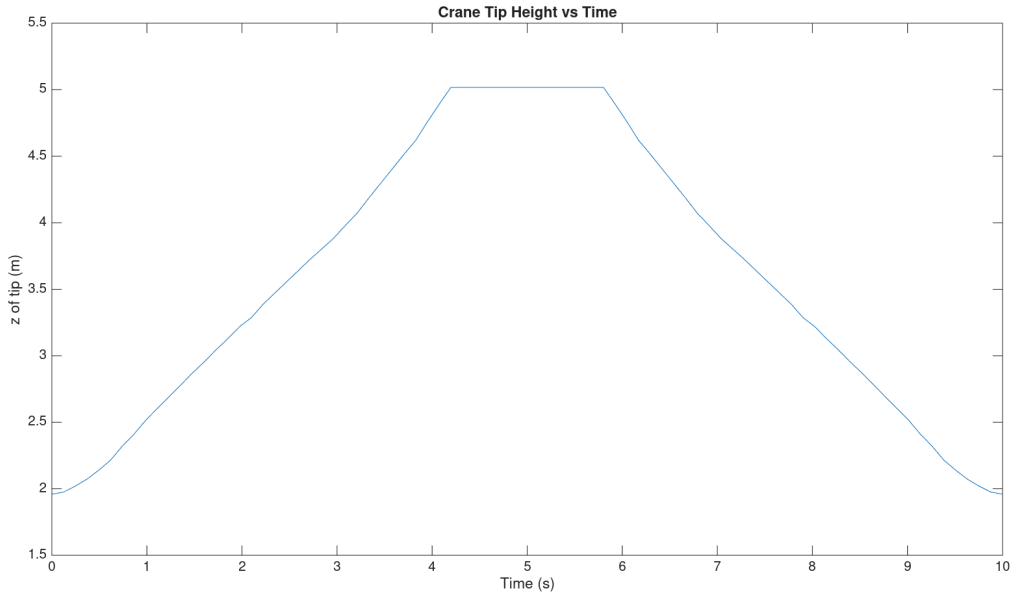


Figure 1: Height of crane tip with respect to time

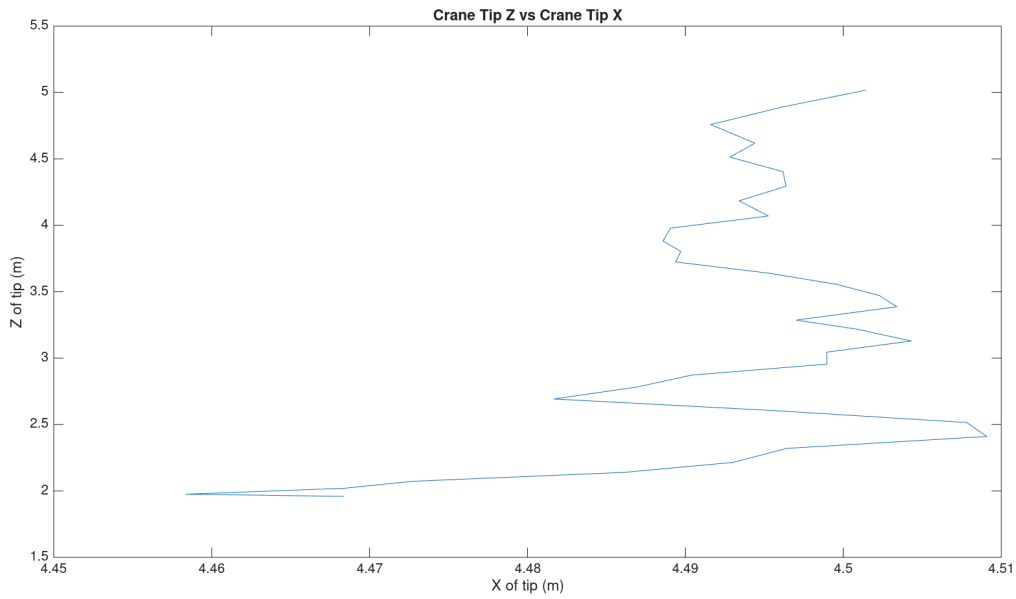


Figure 2: Crane tip position in space during drive cycle

2.2. Cylinder Drive-cycle

From here we continued by determining the forces on the two cylinders throughout the drive-cycle. We did this by calculating the moment generated on either arm hinge by the respective cylinder and the mass at the crane tip. We determined the resulting forces to have the equations:

$$F_1 = M_t \cdot g \cdot \frac{(L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_2))}{e_1}$$
$$F_2 = M_t \cdot g \cdot \frac{L_2 \cdot \cos(\theta_2)}{e_2}$$

The equivalent MATLAB code is as follows:

```
F_1 = M_t*g*(L_1*cos(th_1)+L_2*cos(th_2))./e_1;  
F_2 = M_t*g*L_2*cos(th_2)./e_2;
```

With F_1 and F_2 determined, we could now plot how they changed through time.

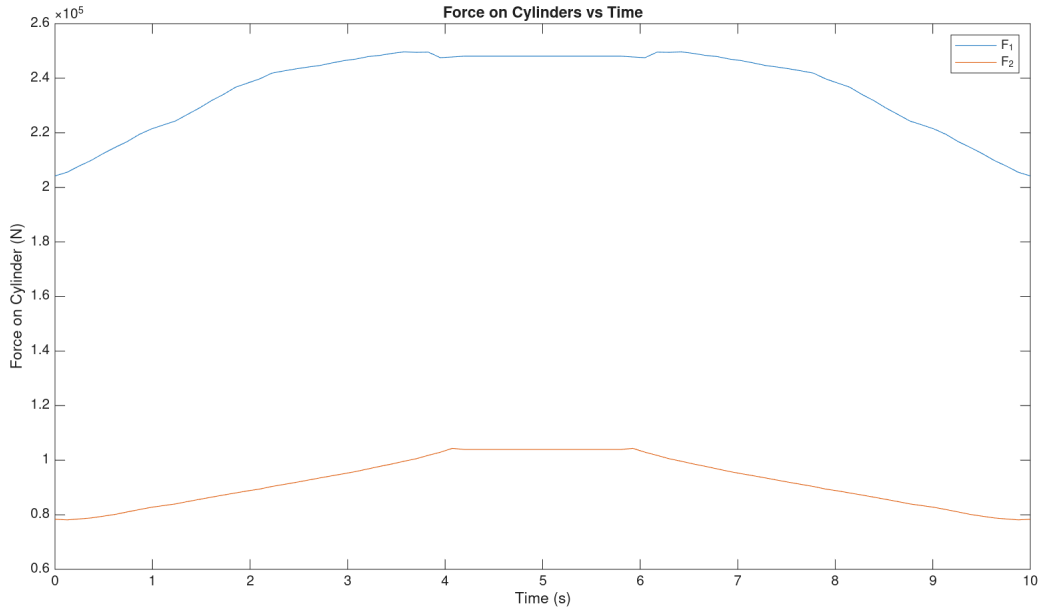


Figure 3: Forces on the two cylinders in the system

2.3. Worst-case Scenario and Cylinder Diameter

Using the same equations we used to determine the forces during the drive-cycle, we can find the forces in the worst-case scenario for the system, with the MATLAB code as follows:

```
F_wc_1 = M_tmax*g*(L_1*cos(th_1_wc_cyl1)+L_2*cos(th_2_wc_cyl1))./e_1_wc_cyl1;  
F_wc_2 = M_tmax*g*L_2*cos(th_2_wc_cyl2)./e_2_wc_cyl2;
```

In this case the θ and e values for either cylinder during the drive-cycle were replaced by the given worst-case scenario values. From these forces we could determine the diameter of the cylinder required to withstand these inopportune forces. We calculated the area and diameter required as:

$$A_1 = \frac{F_{1,wc}}{\eta_c \cdot (p_{1,max} - 0.5p_2)}$$

$$D_1 = \sqrt{\frac{4 * A_1}{\pi}}$$

$$A_2 = \frac{F_{2,wc}}{\eta_c \cdot (p_{1,max} - 0.5p_2)}$$

$$D_2 = \sqrt{\frac{4 * A_2}{\pi}}$$

In which $p_{1,max} = p_{s,max} - p_{extra}$, where p_{extra} is the unavoidable losses in the system, which we have to account for and which we arbitrarily determined to be $p_{extra} \approx 2000000 Pa$. We determined the following values for the respective variables:

- $A_1 = 0.020437 \text{ m}^2$
- $D_1 = 0.1613 \text{ m}$
- $A_2 = 0.014064 \text{ m}^2$
- $D_2 = 0.1338 \text{ m}$

Because no manufacturer produces cylinder of every size, we need to determine the cylinder with diameter which fits our needs. To do this we looked through *Bosch Rexroth's* linear hydraulic cylinder catalogue, and found a suitable candidate. Out of the catalogue we ended up choosing a two Series CGH3 type hydraulic cylinders. These cylinders are mill-type, double acting and can withstand the requirement maximum pressure. Of the CGH3 cylinder options we chose a cylinder with a diameter of 180mm and for the second cylinder a diameter of 140mm. In total the following data was determined necessary:

- Cylinder 1 Bore Diameter: $D_1 = 0.18 \text{ m}$
- Cylinder 2 Bore Diameter: $D_2 = 0.14 \text{ m}$
- Cylinder 1 Area on Piston Side: $A_{p1} = 0.025447 \text{ m}^2$
- Cylinder 1 Ring Area on Rod Side: $A_{r1} = 0.013175 \text{ m}^2$
- Cylinder 2 Area on Piston Side: $A_{p2} = 0.015394 \text{ m}^2$
- Cylinder 2 Ring Area on Rod Side: $A_{r2} = 0.007540 \text{ m}^2$

To be absolutely sure we made the right choice we also found the maximum stroke velocities for the respective cylinders and determined them to be:

- $v_{1,max} = 0.14m/s$
- $v_{2,max} = 0.16m/s$

We then plotted these two values alongside the absolute value of the velocities of the cylinders during their drive-cycles.

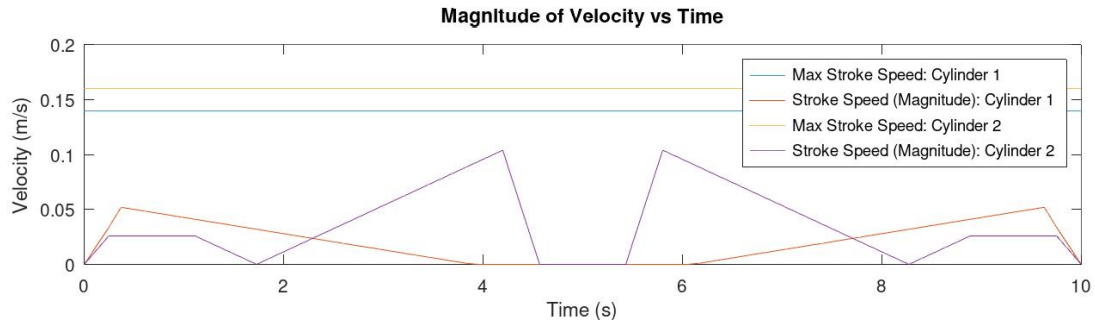


Figure 4: Determination of being under the maximum stroke velocity throughout cycle

We can see that we are under the maximum stroke velocities throughout the drive-cycle, so there is no issue with our choice.

3. Circuit Drive-cycle Analysis

4. Pump Selection

5. System Analysis and Efficiency

6. Cooler Dimensioning